

Recycling of aggregates in the World



Working for sustainable development



Green Mineral

Business plan

NATIONAL STRATEGIC PROJECT ON THE
COLLECTION OF DEMOLITION RUBBLE
AND GLASS AS A MODEL OF CIRCULAR
ECONOMY

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1 BACKGROUND

Project opportunity to date

Green Mineral has proposed a strategic National Circular Economy project based on clear rules and development programs with a twenty-year projection for the National Government of South Africa in order to have national coverage on the selective collection of demolition rubble and glass recycling for all municipalities in South Africa. (read from **page 4** to **page 20** of the Dossier presented to the **Ministry of the Environment with MCE protocol 211993 of November 2021**). The project has many advantages for the municipalities, such as the elimination of the proliferation of illegal landfills of building materials and the creation of jobs (all listed on the page 32 index of the Dossier and, again here, for greater clarity, an extract of the benefits **is attached**).

1.1 Brief description of the promoters/sponsors

Currently, the sponsor of the project is an Italian company called **Alseco** (<https://www.alseco.cloud/>). At the request of the Ministry of the Environment, **Alseco** had already made the private share capital available in 2022 (see attachment).

1.2 Company background and shareholders, including group structure

Green Mineral's background is firmly rooted in Italy, which is the country of origin of the company's founder. The company, NCL Group (<http://www.nclgroup.it/en/index.html>), of which the undersigned is a partner, has been operating in the waste recovery sector since 1998 in accordance with the release of the Ronchi decree (<https://www.ciclia.it/decreto-ronchi/>) and also following 2008/98/CE Directive. This Directive served to establish the objectives of collecting 70% of inert waste by 2020 (see attachment).

1.3 Business strategy

The company's business model is basically to assist municipalities that currently have great difficulty in managing the flow of demolition debris. This is accomplished with the recovery and treatment of demolition waste and glass in dedicated plants on municipal areas or industrial lots that would be assigned to us free of charge. It would then be possible to produce various products to either sell in the construction sector or utilize for infrastructure work. This is all in order to eliminate the proliferation of new landfills and to remediate the existing landfills. This is of benefit to the municipalities and the environment. These remediation costs will not impact the pockets of citizens in the future (see news extract from the web).

1.4 Objectives of the project's scope

The program's objectives are to help municipalities address the problem of construction, demolition and glass waste, which they currently have to deal with. In turn, this promotes an aesthetically pleasing environment and a better place to live for everyone. It also creates employment. To achieve this type of result, it is necessary to work on the Municipal regulations in order to control the flows of demolition waste in order to eliminate the proliferation of illegal landfills, today, **from the findings on the Web, a great plague in the country and costs for the clean-up that will fall on the community**

2 PROJECT DESCRIPTION

2.1 Project History and Current Status

Green Mineral has been established on a solid foundation in Italy. After careful analysis, various solutions have been developed for the country and the region. These solutions serve to help the human race to safeguard the environment with well-formatted and updated technology. This would ultimately benefit Africa and the rest of the world. In the introductory report on page 3 of the Dossier, which is registered at the Ministry of Environment, entitled MCE 211993 of 17 November 2021 (see attachment). The strategic project and its potential operational benefits are briefly described in the attachment.

2.2 Project Location

The project sites would include all the municipalities of South Africa, starting from the 22 identified in the Green Mineral Dossier (see pages 4 and 5). For this reason it will take no less than 15 years to complete the National program. Considering that the tariff on demolition rubble been noted only in some municipalities, the first plants could be installed in Johannesburg and Durban.

The reasons for starting with the two municipalities are dictated by the fact that there is a specific tariff on demolition rubble.

The size of the industrial lots varies from **15.000 to 30.000 square meters**, depending on the three phases of work to be carried out.

The industrial sites will be requested free of charge from the municipalities in exchange for concessions. In the event that the municipalities are not available to grant the lots, they would purchase the lots to be utilized.

Since these sites would be located in industrial areas (even abandoned lots could be utilized), they will need to have water, sewer and electric service.

2.3 Project Structure

For each identified municipality, the future Green Mineral South Africa company (composed of two private entities: **Green Mineral and Alseco**) and the IDC (the public sector), **and even better if the municipal company Pikitup is also involved** will build the first recycling and other plants described in phases 1- 2 and 3 with collection centers (**see page 4 of the Dossier**). The descriptions of the various activities are all reported in the power point presentation that we will send to you in a separate email.

2.4 Project Partners

Currently, **Alseco** (an Italian company) is the financial partner to the project.

The project is currently owned by the Italian company Green Mineral, thanks to its industrial patent for Zero process waste (**see page 21 of the Dossier**). While the **company Alseco**, in addition to being a financial partner, is a leader in Italy in the recovery of other waste such as rubber and plastic.

2.5 Technical

Green Mineral is technically strong thanks to the technological progress of the country. We are the first in the world to recover processing waste and process it later in phase 3.

The capacities of a typical plant are well specified in the index page of the power point that we will send you with another attachment.

The flow chart is well specified in the index page of the power point that we will send you with another attachment.

The description of the process is well specified in the index page of the power point that we will send you with another attachment.

The mass balance is well detailed in the index page of the power point that we will send you with another attachment.

The raw materials and their requirement are well specified in the index page of the power point that we will send you with another attachment.

The specifications and origin of the raw materials are well detailed in the index page of the power point that we will send you with another attachment.

The types of granulates produced are well specified in the index page of the power point that we will send you with another attachment.

2.6 Potential Market

The potential market of Green Mineral South Africa is huge with quantities of demolition rubble (C & D) collection estimated at around **10.000.000 million** tons of

annual collection over the next 15 years and without considering the share of glass where we have not been able to gather information.

Currently the size of the national market is close to zero, little is collected and everything goes to landfill, except small entrepreneurs with small mobile crushing plants.

As regards supply and demand, the market is yet to be built but is promising. There is a tax on 2 municipalities and therefore the waste delivered to the plants will have to be paid. As well as recycled granulates will be sold at competitive prices (10-15% less), compared to natural materials from the extraction quarries.

The main customers are all construction companies and private individuals. Even the Municipalities could be potential customers if they manage small direct and urgent works (**for example, a landslide on a road**).

There is no real segmentation of customers, but for private individuals who purchase there will be no discounts for them.

There is no market gap because the entry prices on waste disposal will be regularized throughout the nation, through a law or regulation for each Municipality. The sales prices will be equal to the prices that will be found for each region where there are extraction quarries. With the National or Municipal tax on incoming waste, prices will never be able to change negatively.

The potential reference market is only the construction, residential, industrial and infrastructure sectors.

The players in the sector are always construction companies and private individuals.

The potential sales volumes are dictated by supply and demand and, at full capacity, after 15 years we have estimated revenues of \$ [REDACTED] and profits of \$ [REDACTED] on potential sales volumes of **10.000000 million tons**

But, to ensure that these recycled granulates can be relocated in large public works, it will be necessary to prepare a regulation on the reuse of recycled granulates also in public works, so as to oblige companies that win public tenders to use these products. These procedures have been active throughout Europe for years, including Italy – the link <https://www.mase.gov.it/pagina/riuso-e-preparazione-il-riutilizzo>

2.7 Key Indicative Agreements

in order to activate this National Circular Economy program and get this strategic project off the ground that will provide work for around **7.200 people**, it will be necessary to sign the parasocial agreements (see pages 20 and 35 of the Dossier). Now, since there are 2 large operational municipalities with the costs of demolition rubble, it is much easier to start with them rather than immediately proposing a National tax to the Ministry of Finance. **With the prices of the rubble detected, the project is financially sustainable.**

2.8 Regulatory Requirements

Green Mineral works closely with various local stakeholders in order to ensure compliance with various environmental regulations and municipal by-laws.

In South Africa there are now general rules, but **Green Mineral** proposes to sign parasocial agreements that are highlighted on page 35 of the Dossier, having as its object Draft Convention. The signing of this document will be able to guarantee the final result of the strategic project that will modify both the municipal regulations and the reuse of recycled granulates in public works with a specific circular from the competent Ministry.

2.9 Developmental Impact

The impact on development is beyond measure. Looking at the current state of the municipalities in the country, they struggle with infrastructure issues, such as dealing with the waste we seek to treat. Municipalities do not have an answer to the problem and their airspace is quickly filling up.

The impact on development is immeasurable. Considering the current state of the country's municipalities, they are now struggling with management problems, such as channeling this waste now to authorized landfills. While, with the construction of dedicated plants (about 400 for the 3 phases of work) there will be an answer to the problem and the future will be bright, that is, better managing the environment around us and giving work to many people, about **7.200** (see page 20 of the Dossier).

2.10 Critical Success Factors

Municipalities will benefit greatly from the proposed solution. This solution is what municipalities across the country are in great need of. Some have closed landfills because they could not recycle waste. If this is resolved, those landfills would be open.

With the control of flows following the changes in municipal regulations, the illegal landfills that are proliferating across the country today will be eliminated. See link Mapping of illegal landfills in September 2023 <https://www.mdpi.com/1660-4601/20/18/6750>.

3 PROJECT RATIONALE

The project tries to address the problems faced by municipalities regarding **C&D** and tries to solve the issues. Waste in municipal landfills takes up a lot of space. The project makes the environment safer and introduces reuse and recycling solutions. All this is schematically reported in the attached sheet” **direct and indirect benefits for municipalities**.

4 COMPETITIVE ADVANTAGES

Green Mineral is bringing cutting edge technology to address the problems. **Green Mineral** possess first world skill and experience. Plants will be built with zero waste processing processes. See process patent on page 21 of the Dossier.

5 RISKS & MITIGANTS

The risks observed are those associated with rigid government policies, obsolete methods and slow adoption of change. Our mitigation solutions imply the involvement of municipalities that will have to make themselves available, both **for the concession of industrial areas**, also to be redeveloped and in modifying the regulations and then all the ministries that will have to give the authorizations to carry out industrial activities and have a direct relationship in dialogue with the various managers to briefly aim for a better and sustainable future.

6 SALIENT FIGURES

The main feature of the project is that municipalities face a major problem with landfills. People migrate to cities and then settle there permanently to improve their lives. Ultimately, this creates new challenges for basic services, such as the need to dispose of the waste generated by these families and businesses. This directly creates challenges for landfills, which are difficult to manage both economically (**realization costs**) and environmentally (complicated management of the same that often leads to pollution of the groundwater) and the need to deal with the waste that continues to accumulate in landfills. Green Mineral has excellent solutions to help municipalities address the problem, which is to convey this waste to recovery plants. This generates 2 positive factors:

- an extension of the life of landfills
- a reduction in the purchase of raw materials from quarries with the protection of the environment

7 STRATEGIC FIT

If the **public company IDC** firmly believes and supports the Green Mineral supply chain initiatives as a model of 'Circular Economy', as they influence various aspects of the economy,

including the creation of jobs within the supply chain. We therefore invite **IDC** to speak directly to the heart of Green Mineral, not to mention the advantages in the possibility of reclaiming the illegal landfills that currently exist throughout the country. In addition, the environment benefits greatly and above all the citizens who will not be involved with new taxes to support the reclamation of existing illegal landfills (see attached findings on the web as of August 30).

8 REQUEST

The request to **IDC** is for co-funding of the **project**. **IDC** has already indicated they do co-funding of projects, such as the one that Green Mineral proposes bringing to the country.

9 ROADMAP, DELIVERABLES AND TIMELINE

Green Mineral would be at peace knowing that its financing needs for its participation in the capital of the newco **Green Mineral South Africa (PPP)** and its needs for the project would be met by the **IDC** which in its regulations provides for a maximum participation in the capital of 49%, and with you we distribute the project in various municipalities starting from Durban and Johannesburg.

10 REFERENCES

The perfect point of reference is the example of what **Green Mineral** is already doing in Italy regarding similar projects since 1998.

The perfect reference point is the example of what Green Mineral is already doing in Italy for similar projects since 1998 and then abroad since 2018, **Ivory Coast, Morocco, Albania, Algeria, Pakistan and Romania.** <https://www.greenmineral.it/en/news/>

Strong experience of its owner. (see CV page 37 of the Dossier). We are currently in contact with the public company RER Ecologic Group SRL <https://rergroup.ro> to which we have proposed a public private partnership PPP

11 ANNEXURES

- Project submitted to the Ministry of the Environment prot MCE211993 of 18 November 2021
- Expression of interest by the Alseco company
- Directive 2008/98/CE
- Direct and indirect benefits for Municipalized Companies
- Costs for the remediation of illegal landfills detected on the web as of 30-August 2024
- Price survey of granulates from quarries in the geographical area of Johannesburg
- Survey of rates on demolition rubble from the municipal company PIKITUP as of 30th August 2024
- Summary of revenues with a twenty-year projection
- Power Point